

Sang-yun Lee (Yun)

Philadelphia, PA, USA | ideaist@engineering.upenn.edu | +1 (267) 679-0962 | yunlee.framer.website

EDUCATION

University of Pennsylvania

Master of Integrated Product Design (M:IPD)

Philadelphia, USA

Aug. 2026 – Present (Expected 2028)

- Interdisciplinary program at the intersection of Penn Engineering, Weitzman School of Design, and Wharton — pursuing technology-driven strategic design with an early focus on healthcare and accessibility applications

Sungkyunkwan University

Master of Service Design (4.3 / 4.5)

Seoul, Korea

Sep. 2022 – Aug. 2025

B.A. in Global Business Administration & B.S. in Data Science (3.95 / 4.5)

Mar. 2017 – Feb. 2022

- Collaborated on strategic service design, CX improvement, service planning, and branding strategy projects with Samsung, Ministry of Culture, Sports and Tourism, Samsung Seoul Hospital, etc.
- Thesis: *Data-driven Disability Persona Augmentation for Service Journey Inference* — Focusing on Daily Living Support Needs of People with Developmental Disabilities

WORK EXPERIENCE

Samsung Electronics | Digital Appliances Division

Business Developer, Market Intelligence Team

Suwon, Korea

Jul. 2025 – Jul. 2026

- Identified business opportunities through customer usage data analysis, demand forecast, and market sensing; orchestrated global market demand analysis using GfK, Euromonitor, and BSRIA historical data (2017–2025), producing 5-year-ahead demand projections through 2030
- Built an enterprise-wide demand dashboard and AI chatbot via vibe coding (AX project), aggregating and reconciling demand data across CMI, sales organizations (retail, subscription, online, B2B), and regional sales (RPM); presented prototypes to sales leadership and executives for evaluation
- Proposed a Bixby-powered cooking-assistant service for screen-based AI-appliances (Family Hub refrigerators), enabling hands-free voice interaction while cooking; adopted as an official product development initiative

SDI Lab | Data-Driven Service Design Business Strategy Consulting Institute

Researcher

Suwon, Korea

Jul. 2024 – Jul. 2025

- Conducted data-driven service design research and consulting projects across public-sector, corporate, and healthcare clients, quantifying personas from behavioral and usage data and deriving service strategy from them (see Projects)
- Led disability- and aging-focused service design research, defining user personas from real behavioral data and using AI-driven simulation to experiment with service strategies against them (see Publications)
- Co-organized and ran large-scale, cross-institutional student programs (100+ participants), mentoring students on design thinking methods (see Projects)

ROKA | Republic of Korea Army

Headquarters Company Commander, Forward Observer Officer

Injae, Korea

Mar. 2022 – Jun. 2024

- As the youngest Headquarters Company Commander, coordinated and directed 60+ personnel across six functional sections — transportation, ammunition, communications, medical, operations, and intelligence — as the cross-functional program lead for all non-combat support
- Served as a Forward Observer Officer at a front-line Guard Post, maintaining operational stability during a North Korean missile provocation (Oct 2022)

BCG | Boston Consulting Group

Research Analyst

Seoul, Korea

Oct. 2021 – Dec. 2021

- Contributed to a strategic market-entry analysis for a leading Korean electronics manufacturer's automotive camera business, deconstructing the component value chain and competitive landscape to identify build-vs-buy opportunities and propose M&A targets

Mirae Asset Daewoo

IB, Corporate Finance Division Intern

Seoul, Korea

Jan. 2019 – Feb. 2019

- Communicated with IR officers, contributed to IPO efforts, and prepared an analysis report on the domestic aviation market and LCC, scored highest in overall evaluation at the end of internship

PROJECTS

SDI Lab | Data-Driven Service Design Projects

Since undergraduate research participation

- **National Rehabilitation Center:** Led daily-living journey-map research with NRC researchers across individuals with spinal cord injury, developmental disabilities, and stroke; identified the gap between research insight and actionable solutions, then bridged it by conceiving and directing a 40-participant cross-institutional hackathon with SKKU to translate the research into prototyped interventions
- **VIP Concierge Robot Service:** Designed a robotic VIP-reception service for Sungkyunkwan University's President's Office inspired by Korea's ancient royal court protocol, orchestrating 3 robots into interlocking personas — welcome, guided tour and explanation, and refreshment service — and coordinated with Mechanical Engineering robotics researchers to deploy all 3
- **SAIHST** (Samsung Advanced Institute of Health Science & Technology): Participated in a welfare & rehabilitation service design

project for hospitalized patients using the metaverse, designing the UI/UX for planned welfare services in a hospital super app and prototyping a collaborative artwork service using Unity and Magma

- **AIEX** (Applied Innovation Experience Design Track): Planned & organized a 118-student design-talent training program, including the NRC joint hackathon, a design thinking school for building startups with Figma training, a published conference paper on the outcomes, and a capstone project with Tokyo Polytechnic University
- **Shinsegae Starfield**: Conducted a customer demographic analysis for mega-shopping mall branding, planned F&B services and designed overall service for newly opening malls
- **Ministry of Culture, Sports and Tourism**: Managed and experimented digital public communication strategy, selected as an OECD Best Practice
- **Samsung Electronics**: Completed a brand fandom analysis project and a refrigerator service design with the Household Appliance Division, as well as a virtual urban service design project using the metaverse with Raemian

Book Publishing

2023 Social Trends: Universal Design

Mar. 2023

- Introduction to universal design and a shift from welfare to market-based approach for disability

PUBLICATIONS

Data-driven Disability Persona Augmentation for Service Journey Inference: Focusing on Daily Living Support Needs of People with Developmental Disabilities

Published, Sungkyunkwan University

Jun. 2025

- Augmented Vineland-based adaptive behavior data from 9 individuals with developmental disabilities to a dataset of 2,400 cases, addressing the small-sample constraint that typically blocks data-driven disability research
- Derived 5 representative developmental disability persona clusters via multidimensional analysis
- Implemented AI-driven personas and ran ADL/IADL scenario simulations to quantify support needs, providing a reusable framework for service design in low-data disability domains

Patterning Military Experience for Quality Improvement of Enlisted Soldiers: A Persona-Based Study on Military Life Cycle Experience Design

Published, Korean Academy of Military Social Welfare

Dec. 2024

- Interviewed 18 high-performing soldiers to quantitatively measure their military service experiences, deriving a 5-stage life-cycle pattern model (Adaptation, Growth, Maturity, Stagnation, Decline) to inform predictive HR programming for new recruits

EXTRA-CURRICULAR ACTIVITY

Federation of Korea ROTC 60th Officer

Seoul, Korea

President

Mar. 2022 – Present

- Represent 3,561 ROTC 60th officers; organized a job fair with Shinhan Financial Group, promoted veteran recruitment with KB Kookmin Bank and E-Land. Partnered with 10 hospitals for their families

DeepBridge | Regular Customer Management AI Tool Start-up

Seoul, Korea

President

May. 2021 – Oct. 2021

- Customer mgmt software startup using POS and CCTV data, previously registered in SKKU campus town

GRU | Start-up Strategic Consulting Club

Seoul, Korea

President

Jan. 2020 – May. 2021

- Led design thinking education and established HR system, completed consulting projects with diverse startups in education, beauty, proptech, digital marketing, tourism, personal mobility and smart factory
- Collaborated with SDI Lab, managed projects with Ministry of Culture, Sports, and Tourism, and Samsung

AWARDS & HONORS

NEO The MacGyver Award, Samsung Electronics (Aug. 2025) | KD Pharma-Songcheon Foundation Scholarship, x2 (2024–2025) | GP Border Operation Merit Award, ROKA (Mar. 2023) | 3rd Prize, Mirae Asset Financial Big Data Festival (Nov. 2021) | Yoon Yong-taek SKKU Alumni Chairman's Scholarship, x2 & 2nd Prize, 14th Convergence Foundation Project (2020–2021)

SKILLS

Tools & Software: MS Office (PowerPoint, Excel), Rapid Prototyping, Vibe Coding, Figma, Unity, Magma, Rhino, Onshape

Languages: Korean (Native), English (Fluent), Chinese (Conversational)